

Product Requirements Document

Functional and Non-Functional Requirements

1. Actors

Role	Responsibilities
Dispatcher / Operations Engineer	Monitor services, investigate incidents, use AI copilot, approve recommendations
Service Manager	Review customer/SLA impact, approve escalations, inspect analytics
Administrator	Manage users, roles, configuration and audit access

2. Functional requirements

- Authenticate users and enforce RBAC.
- Create/manage customers, applications, services and instances.
- Ingest and query service telemetry.
- Track SLA targets and service health.
- Detect deterministic threshold/SLA violations.
- Create and manage incidents.
- Generate ML risk assessments.
- Search approved runbooks using semantic retrieval.
- Ask AI questions against operational evidence.
- Display evidence and recommendation rationale.
- Allow approve/reject/modify actions.
- Persist AI/tool/audit history.

3. Non-functional requirements

- Security: least privilege, encrypted transport, secret management.
- Availability: backend and database designed for restart/recovery.
- Performance: paginated APIs and bounded AI request timeouts.
- Scalability: horizontally scalable Java/API and asynchronous telemetry path.
- Observability: logs, metrics, traces and correlation IDs.
- Maintainability: modular Java features and isolated Python service.
- Auditability: every high-impact AI decision is traceable.

4. MVP exclusions

- Autonomous infrastructure mutation.
- Full AIOps replacement.
- Unbounded arbitrary SQL by the LLM.
- Complex multi-region deployment.
- Real customer production integrations before the core workflow is validated.